

Sterile Filtration Validation Report

Illustrative risk-based validation structure for a sterilizing-grade filtration process.

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Prepared for	Public ProcessQuill portfolio

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1. Purpose and scope

This sample report demonstrates a risk-based sterile-filtration validation structure for an aqueous parenteral solution using a 0.22 µm sterilizing-grade filter. The report combines filter suitability, integrity testing and process-operating evidence into a lifecycle conclusion.

2. Process boundary

- Bulk solution after final compounding and pre-filtration controls.
- Sterilizing-grade filter assembly and housing.
- Pre-use integrity test where required by the approved process strategy.
- Filtration under monitored differential pressure and temperature.
- Post-use integrity test and aseptic transfer to filling.

3. Validation evidence matrix

Evidence area	What is demonstrated	Example conclusion
Filter compatibility	Product/filter contact under defined time and temperature	No adverse impact on filter or product attributes
Bacterial retention	Supplier/site validation using appropriate challenge organism and worst-case conditions	Sterilizing-grade retention demonstrated
Integrity correlation	Validated relationship between integrity test and microbial-retention performance	Integrity limit scientifically justified
Process envelope	Pressure, flow, time, temperature, filter area	Defined operating range maintained
Pre/post integrity	Executed integrity results	Pass before/after use according to strategy

4. Example validation runs

Run	Max ΔP	Temperature	Filter contact	Integrity pre/post	Assessment
PV-01	0.9 bar	21 °C	2.8 h	Pass / Pass	Accepted
PV-02	1.2 bar	23 °C	3.4 h	Pass / Pass	Accepted
PV-03	1.6 bar	24 °C	3.8 h	Pass / Pass	Accepted

5. Example process limits

For demonstration only, this sample uses an operating differential-pressure limit of ≤ 1.8 bar, a filtration temperature range of 18-25 °C and a defined maximum filter-contact time. These values are not recommendations and must be derived from the actual product, filter, process and validation evidence.

6. Validation conclusion

The fictional evidence supports the defined sterile-filtration process within its approved operating envelope. Routine use remains dependent on approved filter configuration, integrity testing, batch-specific process records, hold-time controls, deviation management and continued supplier/change assessment.